
MAYO,

BUILDING BRIDGES AND SCHEMES: THE STORY OF MAYO AND A THRIVING TEAM

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Multivariate Quadratic (MQ) cryptography is based on the assumed hardness of finding a solution to a **system of multivariate quadratic equations** (over a **finite field**). This problem is called the *MQ problem*.

Known to be *NP-hard*: seems to be hard on average for an extensive range of parameters:

The current record *mod 31* is solving a system of 22 equations in 22 variables.

$$\begin{aligned}x + 5x^2 + 3xy &= 4 \mod 7 \\ x^2 + 5xy + 5y^2 &= 1 \mod 7\end{aligned}$$

General notation

- $n \in \mathbb{N}$ be the number of variables
- $m \in \mathbb{N}$ the number of equations
- For a prime power q , we denote by F_q the finite field of order q

Multivariate Quadratic Polynomials

A system of m (or k) equations (f_k) in n variables in $\mathbf{F}_q$

$$f_k(x_1, x_2, \dots, x_n) = \sum_{1 \leq i \leq j \leq n} a_{i,j}^{(k)} x_i x_j + \sum_{1 \leq i \leq n} b_i^{(k)} x_i + c^{(k)}$$

where:

- First term consists of the quadratic terms with $a_{i,j}$ as the coefficient
- Second term consists of the linear terms, with b_i as the coefficient
- c is the constant term
- All the coefficients are in $\mathbf{F}_q$

$$\begin{cases} y_1 &= p_1(x_1, \dots, x_n) \\ y_2 &= p_2(x_1, \dots, x_n) \\ &\vdots \\ y_m &= p_m(x_1, \dots, x_n) \end{cases}$$

Multivariate quadratic (MQ) problem

$$\mathcal{P} : F_n \rightarrow F_m : x \rightarrow \mathcal{P}(x) = (p_1(x), \dots, p_m(x))$$

The MQ problem asks:

given a multivariate quadratic map $P : F_q^n \rightarrow F_q^m$ over a finite field F_q and a target $\mathbf{t} \in F_q^m$ to find a solution $\mathbf{s}$ such that $\mathbf{P}(\mathbf{s}) = \mathbf{t}$.

or

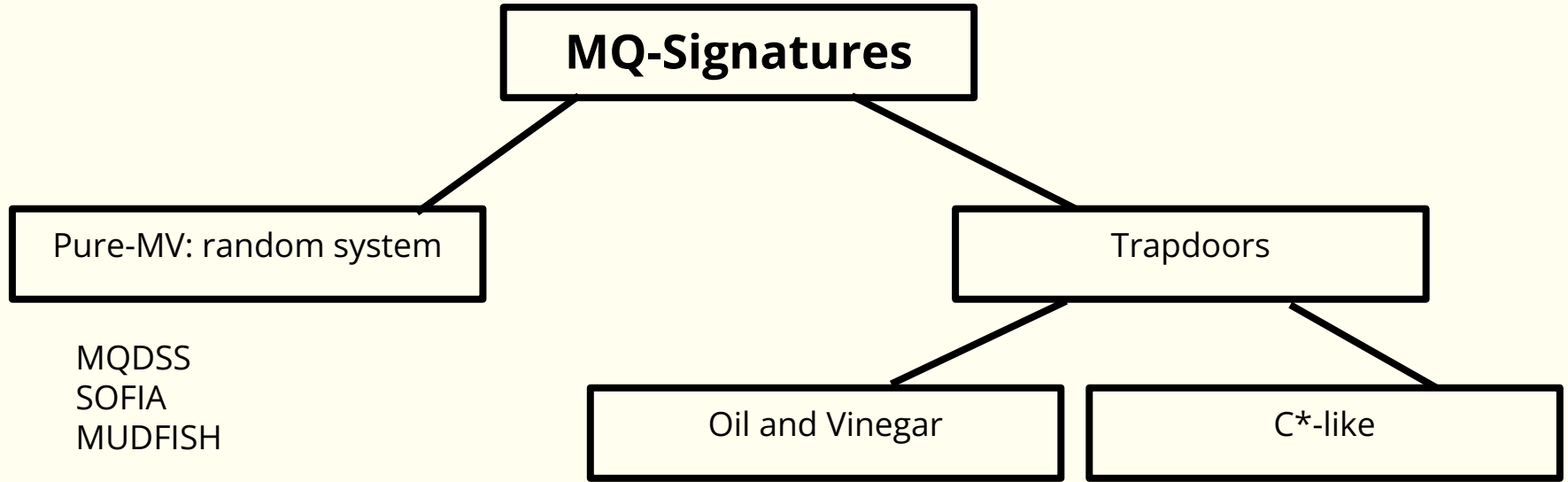
given $(y_1, \dots, y_m)$ in F_q^m , find $(x_1, \dots, x_n)$ in F_q^n with $f_k(x_1, \dots, x_n) = y_k$ for $1 \leq k \leq m$ if they exist.

Multivariate quadratic (MQ) assumption

- Can be a decisional problem: does the preimage (the input) exist?
- We are mostly interested in the **computational version**
- It is known to be NP-hard [1] over any finite field F
- If $n > m(m + 1)$ (underdetermined) or $m > n(n - 1)/2^q$ (overdetermined) the problem can be solved in polynomial time [2] over fields of even characteristic, but when $n \sim m$ the problem is believed to be exponentially hard on average (even for quantum algorithms)
- Best algorithms to solve: $F4/F5$ or XL that use a Gröbner-basis-like approach
 - Ongoing efforts on improving specially for ideals that are globally invariant under the action of a finite abelian group

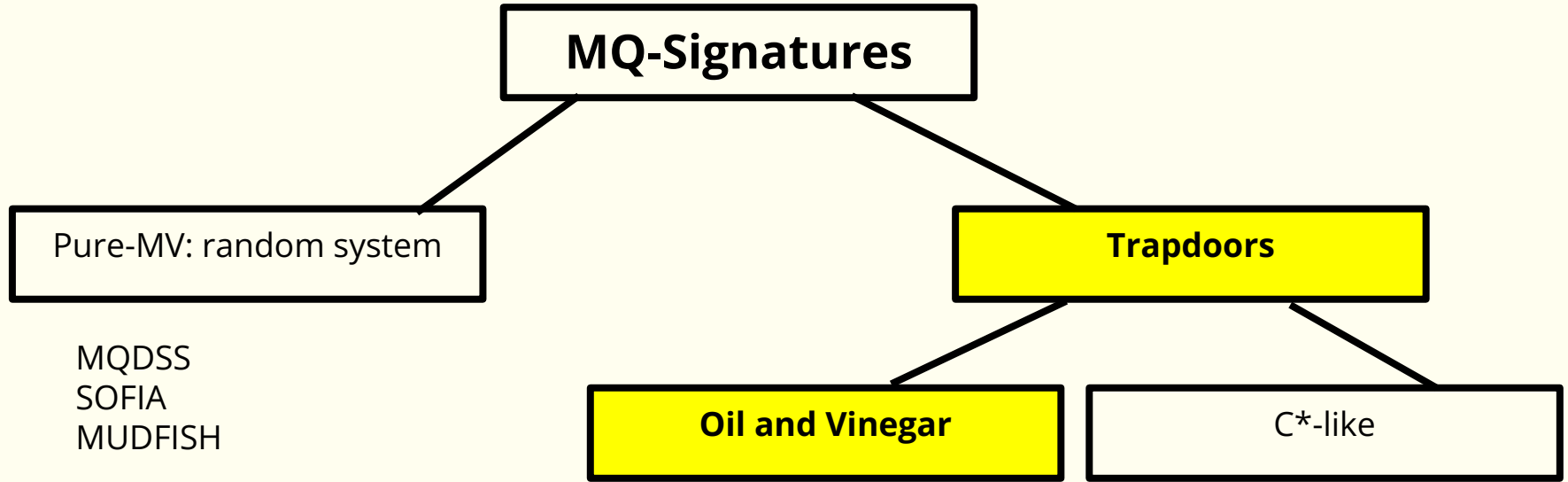
[1] Garey, M. R. *Computers and intractability: A guide to the theory of np-completeness*. Revista Da Escola De Enfermagem Da USP (1979). [2] Thomae, E., and Wolf, C. Solving underdetermined systems of multivariate quadratic equations revisited. In PKC 2012 (May 2012), M. Fischlin, J. Buchmann, and M. Manulis, Eds., vol. 7293 of LNCS, Springer, Heidelberg, pp. 156–171.

Signatures?



Pure-MV: random system: slower, large signature sizes and smaller keys

Signatures?



Trapdoor-based

Trapdoor-based:

- Maps look random but have a “hidden structure” that allows one (a signer) to compute pre-images
- They are based in the *Hash-and-Sign with Retries* approach [3]
 - Trapdoor is probabilistic
 - Public key is generally not surjective

PK: P , description of the trapdoor function

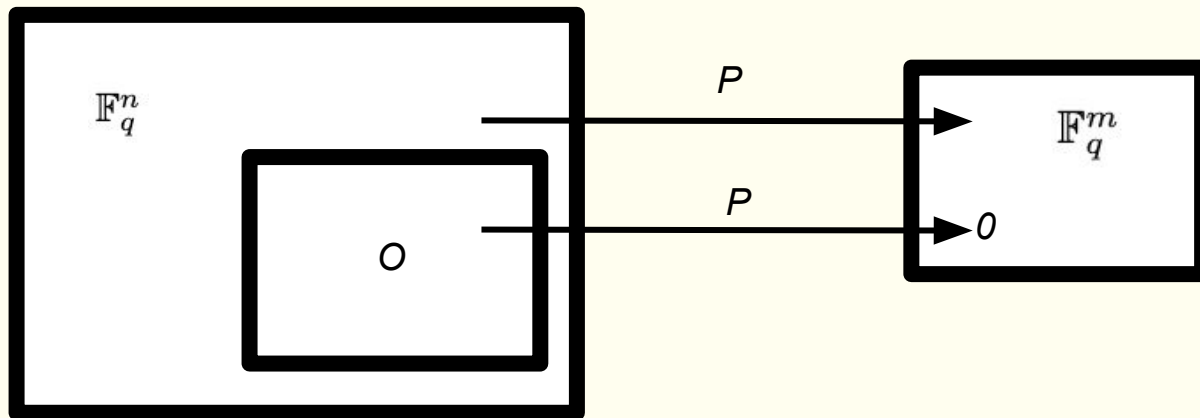
SK: trapdoor information

Signature: input s from $P(s) = H(m \parallel salt)$

```
proc sign(sk : skey, m : msg) : signature = {  
  var i1, i2, z, salt, sol, s;  
  
  (i1, i2) <- sk;  
  z <$ i1;  
  (* repeat until *)  
  salt <$ dsalt;  
  sol <@ H.get(m, salt);  
  s <- i2 z sol;  
  while (s = None) {  
    salt <$ dsalt;  
    sol <@ H.get(m, salt);  
    s <- i2 z sol;  
  }  
  return (salt, oget s);  
}
```

Trapdoor-based: *Unbalanced Oil and Vinegar*

Trapdoor: linear subspace $O \subset \mathbb{F}_q^n$ (oil space) of dimension m on which P vanishes (for every vector: $P(o) = 0 \ \forall \ o \in O$)



Unbalanced Oil and Vinegar: yet another look

Given this description of the trapdoor, we can simplify the description

We define the differential as:

$$\Delta(x) : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m : y \rightarrow \mathcal{P}(x + y) - \mathcal{P}(x) - \mathcal{P}(y) + \mathcal{P}(0)$$

which is symmetric and bilinear in both x and y (we eliminate the linear and constant terms). It shows the purely **quadratic interactions**.

Each component can be written as a bilinear form: $xGy^\top$, with a symmetric matrix G and evaluates to a single field element

Unbalanced Oil and Vinegar: yet another look

Problem: $P(s) = t$

- Pick random $v \in F_q^n$ (the vinegar)
- Solve for $o \in O$ (the oil), such that $P(v, o) = t$:

$$\mathcal{P}(o + v) = \mathcal{P}(v) + \mathcal{P}(o) + \Delta_v(o) = t$$

$$\mathcal{P}(o + v) = \cancel{\mathcal{P}(v)} + \overset{\text{a constant}}{\cancel{\mathcal{P}(o)}} + \Delta_v(o) = t$$

linear system of m equations in m variables.

If no solution, **retry** for another v .

- Signature: $s = o + v$

Let's make it practical! MAYO

- Make the oil space smaller: $\dim(O) < m$
- Vanishes on space dimension o , and it is “whipped-up”tm to vanish on ko

*Mix the oils and vinegars to get somethingTM
add emulsifiers and you get MAYOTM (kinda of)*

non-polar

polar

polar and
non-polar

polar and
non-polar



The amazing team



THE TWO SIDES OF CRYPTO ENGINEERING



* Love your team, not your scheme. 🐼

The amazing team

- Explicit choice of adding people:
 - Not only “good” at the technical side but also welcoming
 - Encourage people to work with us
 - Provide “internships” to work with us
 - Same logic applies to the threshold team
 - The goal is not “to win” but to create practical crypto that encourages research:
 - We want to have fun
 - We want to learn

- The MAYO logo:
 - Keith Haring
 - Joost Swarte
 - Creative, bold, and accessible



My random walk into cryptography

- My degrees are in classical music and communications/literature. I just got my first CS degree!
 - Incredibly useful for me today:
 - Music: not fear of performing or “staring at the blank page”, practice
 - Literature: communication, organization
- I was always interested in understanding how things work and how to create
 - I hated the “is good for you, is appropriate for you, is what is expected”
 - A very stressful childhood
- Different background, and not a straight line:
 - Incredible difficult at times, but the difficulty has created a lot of growth and experience
- Life has many paths
 - No ideal career progression
 - Freedom to choose → not everyone has it s

Lessons learned

- Find your friends and build community
 - Extend bridges to others
 - Building and fighting does inspire people
- Good collaborations are hard to come by
 - Set boundaries
 - Mentorship
- Allyship matters but having someone that knows your lived experience does too
- Research is funnier with stickers and colours
 - Life can come hard at you, humor helps a lot
- Have hobbies
- Job interviews in the industry are weird

Community matters

Decrypting diversity: Montpellier, 16-20 June, 2025,

now yearly!



Community matters

The worst advice I ever got



*You would be more successful
if you focused on just one thing.*



<https://www.youtube.com/watch?v=2bPQLoLPuk0&t=7s>

My fav papers/thesis

- “Improved cryptanalysis of UOV and Rainbow” by Ward Beullens
- “MAYO: Practical Post-Quantum Signatures from Oil-and-Vinegar Maps” by Ward Beullens
- “An Estimator for the Hardness of the MQ Problem” by Emanuele Bellini, Rusydi H. Makarim, Carlo Sanna, and Javier Verbel
- “Cryptanalysis of the Oil and Vinegar Signature Scheme” by Aviad Kipnis, Adi Shamir
- “A Study of the Security of Unbalanced Oil and Vinegar Signature Schemes” by An Braeken, Christopher Wolf, and Bart Preneel
- “Selecting and Reducing Key Sizes for Multivariate Cryptography” by Albrecht Petzoldt

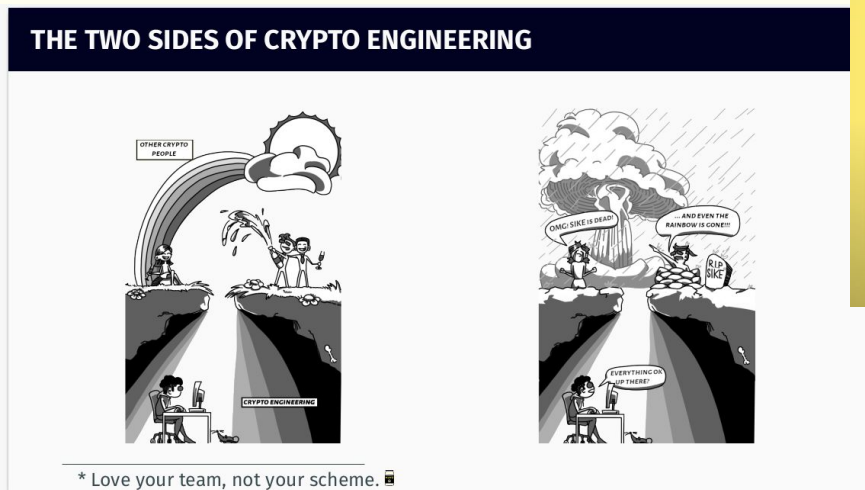
Interested?

- New attacks based on ideals that are globally invariant under the action of a finite abelian group
- Security proofs and expanding for a unified framework of UOV-based
- Formal verification: EasyCrypt or Jasmin
- Advanced constructions: threshold, multi-party and “exotic” signatures



References

- For quick learning: <https://github.com/PQCMayo/MAYO-sage>
- Our C implementation: <https://github.com/PQCMayo/MAYO-C>
- Our website: <https://pqmayo.org/>



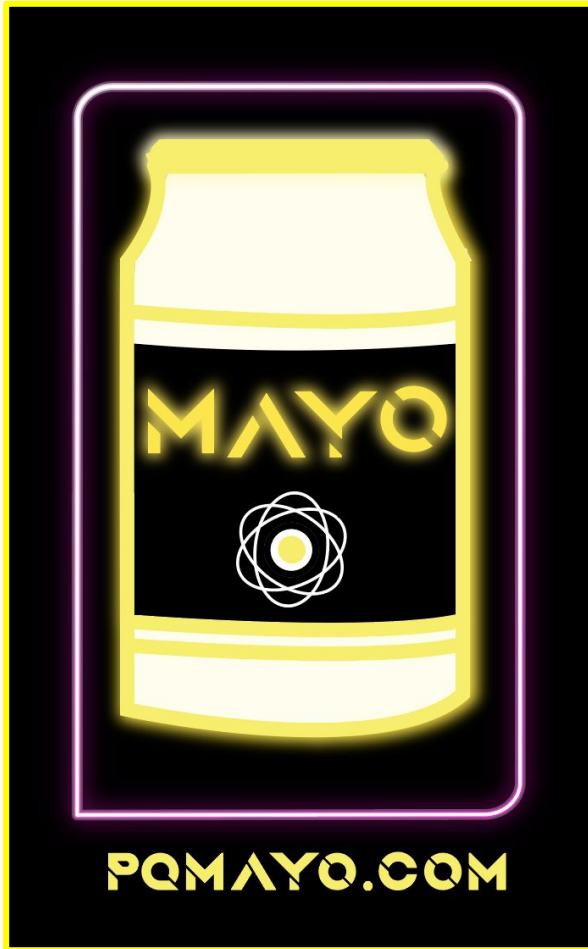
Ask not what MAYO can do for you,
but what you can do for MAYO!

Please reach out to contact@pqmayo.org
if you want to help with:

- Design
- Cryptanalysis
- Implementations
- Security Proofs
- Side-channel security
- Saucy puns
- ...

"90's" version: MAYO'nAES

Logo credit:
Sofia Celi



THANK YOU!

@claucece

<https://sofiaceli.com/>